

Bibliometric Analysis of Digital Finance and Machine Learning Research

Assignment for the Naples Training Week

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Abstract

Digital finance and machine learning have become increasingly intertwined as financial services adopt data-driven technologies and AI-enabled decision-making tools. Despite this rapid growth, there is limited systematic evidence on how research in this domain has evolved over time and diffused across countries. This study provides a bibliometric analysis of approximately 3,500 publications indexed in Scopus between 2011 and 2026, using a structured metadata-based workflow implemented entirely in Python. A targeted search query was used to retrieve literature connecting machine learning concepts with digital finance technologies within a financial context, and the resulting BibTeX dataset was enriched through country extraction. The analysis reveals three main insights. First, publication activity shows a sharp acceleration beginning around 2018, indicating a transition from early exploratory work to a period of rapid expansion. Second, the field exhibits clear geospatial diffusion: research output has broadened beyond traditional academic hubs, with India, China, the United States, and several emerging economies driving much of the recent growth. Third, keyword trends highlight the rising prominence of topics such as deep learning, blockchain, decentralized finance, and AI-based financial applications, indicating technological advances. Overall, the study provides a reproducible and data-driven overview of the evolution of digital finance and machine learning research. The results offer a foundation for more detailed bibliometric analyses and contribute to a broader understanding of how innovation in machine learning is affecting the financial research landscape.

1 Introduction

Over the past two decades, finance has undergone a transformation driven by the rise of digital technologies and advances in machine learning. Developments such as mobile and digital payments, blockchain-based financial systems, robo-advising, algorithmic credit scoring, and algorithmic trading illustrate how both incumbents and emerging actors are reshaping financial services. In parallel, academic research at the intersection of digital finance and machine learning has expanded rapidly.

Despite this growth, there is limited systematic understanding of *how* this research field has evolved. In particular, little is known about: (1) the temporal dynamics of research activity, including the emergence and growth of key topics and technologies, and (2) the geospatial diffusion of digital finance and ML research across countries. A bibliometric study based on metadata

offers a scalable way to map these developments without manually analysing individual papers. This project aims to construct such a dataset and provide an initial overview of the field’s evolution. To address these gaps, this study investigates (i) the temporal evolution of research activity at the intersection of digital finance and machine learning, and (ii) the geospatial diffusion of this research across countries. These questions aim to characterise the development and diffusion of the field in a structured, data-driven manner. The remainder of this paper is structured as follows. In Section 2 we give an overview of the existing literature. Section 3 outlines the data collection procedure, the construction of the bibliometric dataset, and the preprocessing steps applied to extract temporal, geographical, and thematic information from Scopus metadata. Section 4 presents the empirical findings, beginning with the temporal development of publication activity, followed by an analysis of the global diffusion of research across countries, and concluding with a thematic examination of keyword trends. Section 5 summarises the main insights, discusses methodological limitations, and highlights opportunities for future research at the intersection of digital finance and machine learning.

2 Related Work

The existing literature relevant to this study spans several interconnected domains, including digital finance, FinTech, and artificial intelligence in financial applications. This section reviews these strands with the aim of situating the present work within the broader research landscape. We begin by outlining conceptual contributions in digital finance, followed by bibliometric analyses of FinTech research. We then discuss related bibliometric work on AI and machine learning in finance, as well as studies mapping blockchain and crypto-asset research. The section concludes by identifying the gap that motivates our study, the lack of a unified bibliometric analysis capturing the co-evolution of digital finance and machine learning as an integrated field.

Research at the intersection of digital finance and machine learning builds on several streams of prior bibliometric and review studies in FinTech, digital finance, and AI-in-finance. One of the earliest comprehensive overviews of digital finance is provided by Gomber et al. (2017), who introduced the “Digital Finance Cube” to map business functions, technologies, and institutional contexts. Their work illustrated how digitalization evolved from supporting traditional financial services to enabling novel FinTech business models. Subsequent narrative reviews traced similar developments in FinTech innovation and outlined future research agendas, though largely without deploying bibliometric methods. A growing body of work quantitatively maps the FinTech literature. Salem and Shahimi (2025) conducted bibliometric analysis of 2,760 FinTech publications (1968–2025), identifying long-term thematic trajectories and a pronounced surge in research activity. Similarly, Jafri et al. (2025) analyzed 3,884 Scopus-indexed FinTech manuscripts and documented an substantial increase of research post-2015 alongside rising concerns around systemic risk, cybersecurity, and trust. Brika (2022) provided an earlier analysis that highlighted how FinTech and digital finance research co-evolved into distinct thematic clusters. Complementing these broader studies, Li and Xu (2022) examined 1,341 papers on financial innovation, tracing thematic shifts and identifying pandemic-era developments. Bibliometric work has also been applied to niche aspects of FinTech. For example, Bhatt et al. (2023) conducted the first scientometric mapping of cross-technology convergence in e-payment

systems, examining blockchain, AI, mobile technologies, and NFC. Their analysis showed how convergence enhances efficiency but raises interoperability and inclusion challenges. Such studies illustrate how bibliometric findings vary depending on topical scope, data sources (e.g., WoS, Scopus), and analytical techniques.

Parallel to FinTech-focused reviews, scholars have investigated the rise of artificial intelligence in finance through bibliometric approaches. Roy et al. (2025) presented an analysis of 607 AI-in-finance publications. The study identifies major themes such as risk management and market efficiency, and highlighted emerging discussions on ethical AI and AI-driven decision-making. Zakaria et al. (2023) mapped machine learning applications across the financial industry, identifying influential contributions in credit scoring, trading, and fraud detection. Mukthar et al. (2025) examined the AI–FinTech intersection by analyzing work from 2016–2023, revealing rapid growth, influential author networks, and prominent topics such as robo-advising and cybersecurity. Although these studies differ in scope, they converge on the conclusion that AI adoption in finance is accelerating, with unresolved methodological, ethical, and regulatory challenges.

Despite substantial bibliometric work on FinTech and AI-in-finance, no prior study provides a unified bibliometric analysis of the combined development of digital finance and machine learning as an integrated interdisciplinary field. Existing FinTech studies mention AI as one enabling technology, while AI-in-finance studies acknowledge FinTech use cases, but neither stream systematically examines their co-development over time. Furthermore, previous work seldom incorporates spatiotemporal mapping that links research output, collaboration networks, and innovation diffusion across regions. The present study addresses these gaps by offering the first dedicated bibliometric analysis of digital finance and machine learning taken together, leveraging structured metadata (e.g., authorship, affiliations, keywords) to track the field’s global evolution. This provides a longitudinal and geospatial perspective that complements—and extends beyond—the narrower scopes of existing bibliometric studies.

3 Methodology

The analysis in this study is based entirely on bibliometric metadata retrieved from the Scopus database. Scopus was selected because it offers broad coverage of both computer science and finance outlets and provides well-structured metadata suitable for automated processing. The dataset was exported in BibTeX format and imported into Python, where it was converted into a `pandas` DataFrame to enable efficient manipulation of publication years, affiliation strings, and other metadata fields relevant for the analysis.

To identify research situated at the intersection of machine learning, digital finance, and financial applications, a structured Boolean search query was executed in Scopus. The query required each publication to contain at least one machine learning term, one digital finance term, and one finance-domain term within the title, abstract, or keywords. The full query is documented in Appendix A. The search was restricted to English-language journal articles and conference papers published between 2011 and 2026, and the exported metadata included publication year, author keywords, author names, affiliation strings, DOIs, and source titles.

Once imported into Python, the dataset underwent minimal preprocessing. Publication years were parsed as integers to ensure consistency in temporal aggregation. The geographical origin of

each publication was inferred from the affiliation string of the first author. A simple dictionary-based pattern-matching approach was used to detect country names within the affiliation text. Records for which no country could be identified were included in temporal analyses but excluded from geospatial visualisations. No additional filtering, deduplication, or content-based processing was applied beyond the constraints of the initial Scopus query.

For the purposes of constructing the visualisations presented in the results, the cleaned dataset was transformed into three analytical structures. First, a year–country table was created by aggregating publication counts for each country in each year. Second, overall yearly publication counts were computed to examine long-term temporal trends. Third, a keyword–year table was generated by extracting all author keywords, normalising them to lower case, and aggregating their frequencies by year. This structure parallels the country–year table and serves as the basis for analysing thematic developments over time. Together, these programmatically generated tables underpin the time-series plot of annual publications, the geospatial heatmap of country-level diffusion, the stacked bar chart of global research distribution, and the keyword heatmap illustrating the evolution of core topics.

Overall, the methodology follows a reproducible and lightweight workflow: a structured Scopus query, minimal preprocessing focused on temporal, geographical, and keyword metadata, and straightforward aggregation of publication and keyword counts. This approach provides a transparent and scalable foundation for analysing how research in digital finance and machine learning has evolved and diffused over the past fifteen years.

4 Results

This section presents the empirical findings of the bibliometric analysis, structured around the study’s core research questions on temporal and geospatial developments, as well as the thematic evolution of digital finance and machine learning research. The results draw on the cleaned and standardised dataset constructed through the methodology described in the previous section, using aggregated publication counts, country-level distributions, and keyword frequencies to uncover broad patterns in the field’s development. The analysis proceeds in three stages. First, the temporal dynamics of publication activity are examined to identify phases of growth and acceleration. Second, geospatial patterns are explored to assess how research output has diffused across countries over time. Third, keyword trends are analysed to reveal shifts in thematic focus and the emergence of new technological and conceptual topics. Together, these results provide a data-driven overview of how the field has expanded, diversified, and evolved over the past fifteen years.

4.1 Temporal Development of Research Activity

Figure 1 presents the annual number of publications at the intersection of digital finance and machine learning from 2011 to 2026. The overall trend reveals a period of slow and steady growth during the early 2010s, followed by a rapid acceleration beginning around 2018. Publication output increases sharply from 2019 onward, reflecting the surge of interest driven by advances in deep learning, the expansion of fintech applications, and the global rise of blockchain-based

systems. The peak occurs in 2025, after which 2026 shows a decline that is due to studies being indexed to publications in 2026. This temporal pattern suggests that digital finance research has entered a phase of rapid expansion in the last five years, driven by both technological innovation and increased global adoption of data-driven financial systems.

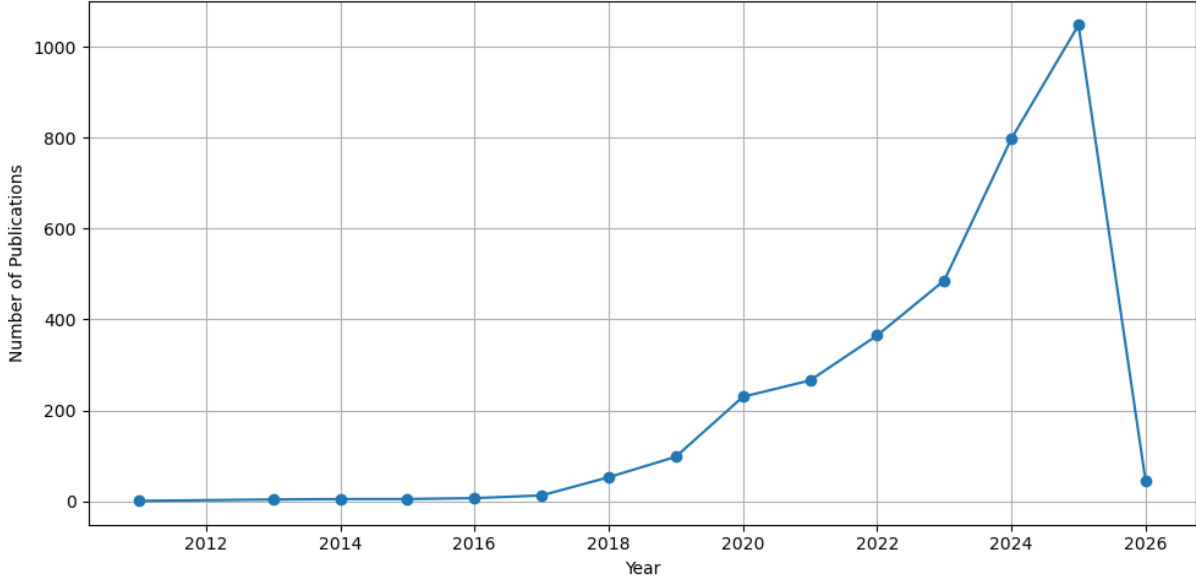


Figure 1: Annual number of publications at the intersection of digital finance and machine learning (2011–2026). The figure shows a period of modest growth during the early 2010s, followed by a sharp acceleration beginning around 2018. Publication output rises steeply from 2019 onward, reflecting increasing academic interest driven by advances in deep learning, the expansion of fintech applications, and the global uptake of blockchain-based financial systems. The apparent decline in 2026 is due to incomplete indexing of publications into the future.

4.2 Geospatial Distribution and Diffusion

To understand how research output is distributed across countries and how this changes over time, two complementary visualisations were generated: a country-by-year heatmap (Figure 2) and a stacked bar chart (Figure 3). The heatmap in Figure 2 reveals a clear diffusion pattern. Prior to 2018, research activity is sparse and limited to a handful of countries. From 2019 onward, output increases dramatically, with India, China, the United States, and the United Kingdom emerging as the dominant contributors. India and China show particularly strong growth, surpassing all other countries after 2020. Other countries such as Indonesia, Saudi Arabia, Malaysia, Italy, and South Korea also increase their participation over time, indicating a broadening geographical engagement with digital finance and machine learning applications.

The stacked bar chart in Figure 3 further illustrates how the composition of global research output evolves. India and China dominate overall volume, while countries such as the United States, United Kingdom, and Indonesia contribute substantially but grow at a more modest rate. The increasing height of the stacked bars over time reflects the global expansion of the field, while the widening distribution across colors indicates diversification in country participation.

Together, these geospatial results show that research on digital finance combined with machine learning is no longer concentrated in traditional academic hubs but has diffused globally,

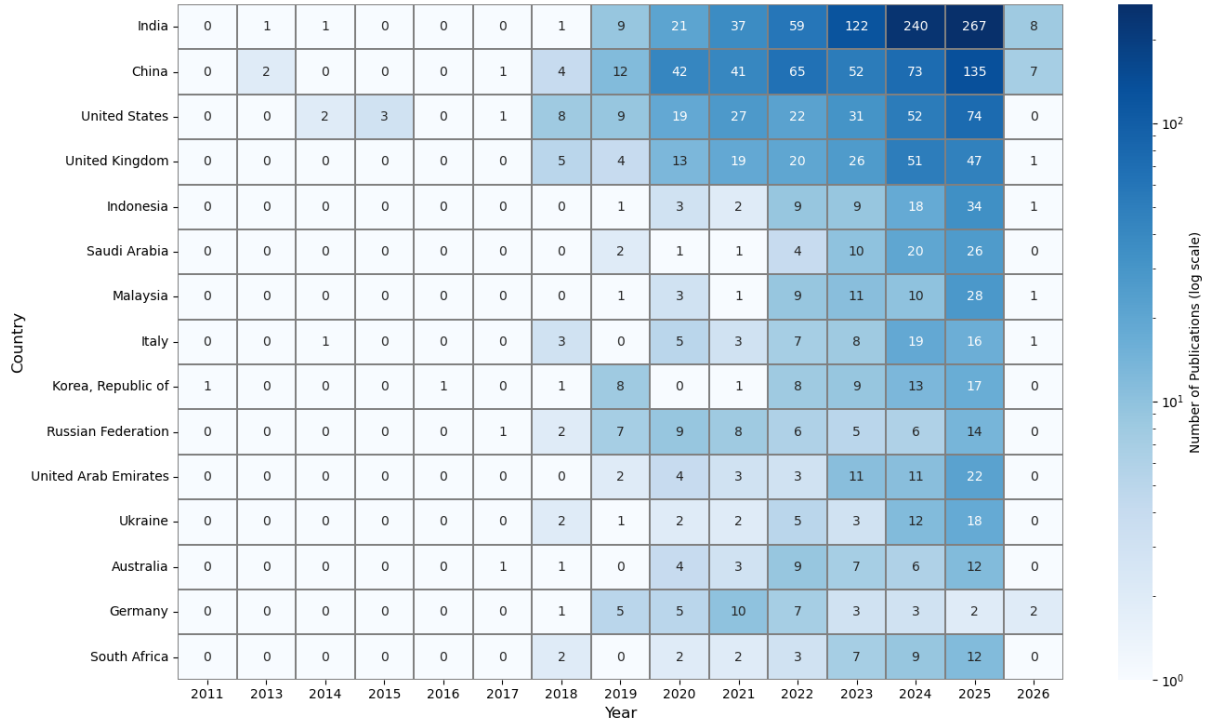


Figure 2: Heatmap of yearly publication counts for the top 15 countries in the dataset (log scale). The figure visualises how research activity at the intersection of digital finance and machine learning has diffused geographically between 2011 and 2026. Prior to 2018, contributions are sparse and concentrated in only a few countries. From 2019 onward, publication output increases sharply, with India, China, the United States, and the United Kingdom emerging as dominant contributors. Several other countries—such as Indonesia, Saudi Arabia, Malaysia, Italy, and South Korea—also exhibit growth over time, reflecting the widening global participation in this research area.

particularly across Asia.

4.3 Keyword Trends and Thematic Development

Figure 4 displays the temporal evolution of the top 15 keywords associated with the dataset, plotted using a logarithmic color scale to account for substantial variation in keyword frequency. The heatmap indicates several important thematic developments. Keywords related to artificial intelligence and machine learning (e.g., *machine learning*, *deep learning*, *artificial intelligence*) begin to rise steadily after 2018, reflecting broader adoption of advanced data-driven methods in finance. Notably, terms such as *blockchain*, *decentralized finance*, and variations of *blockchain* show strong growth after 2019, peaking around 2024–2025, mirroring global adoption of distributed ledger technologies. Traditional finance-related keywords (e.g., *financial markets*, *risk management*) also show increasing usage, suggesting integration of ML techniques into established financial domains. Overall, the thematic analysis reveals a strong pivot toward AI-driven financial innovation and blockchain technologies, with rapid acceleration occurring in the same period that global publication volumes increased.

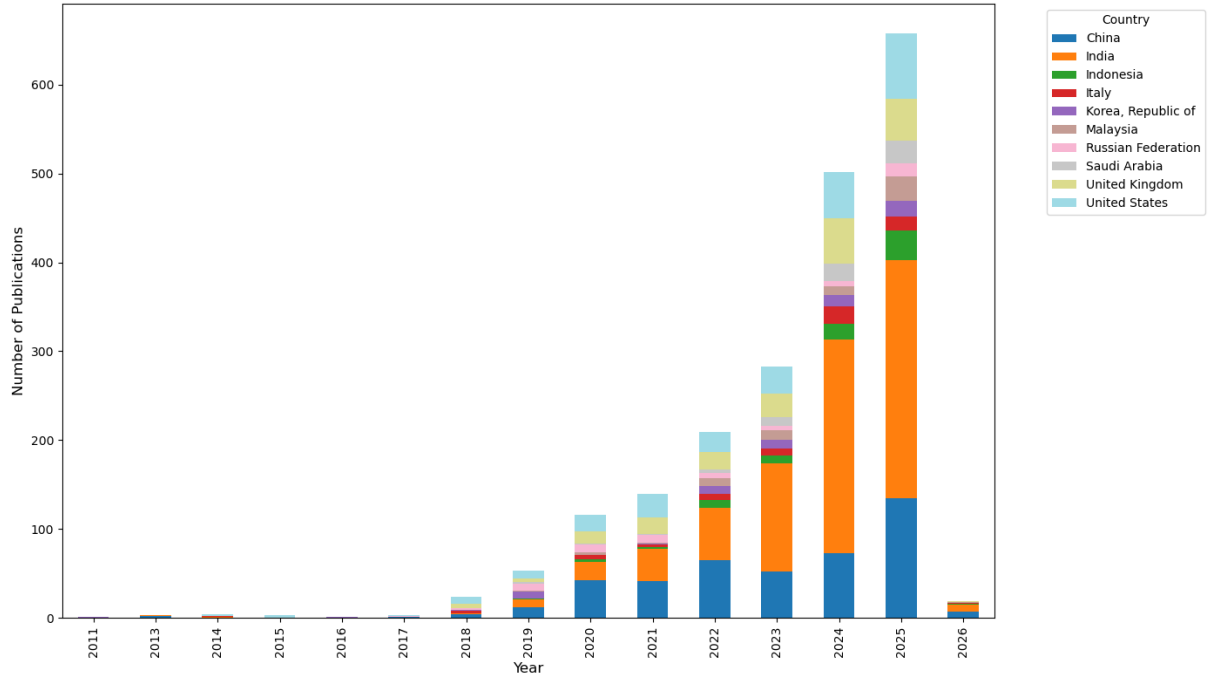


Figure 3: Stacked yearly publication counts for the top ten countries in the dataset. The chart illustrates how the global distribution of research output at the intersection of digital finance and machine learning has evolved between 2011 and 2026. The increasing height of the bars reflects the overall expansion of the field, while the changing composition of the stacks shows a diversification of contributing countries over time. India and China become the predominant contributors after 2020, whereas the United States, the United Kingdom, and Indonesia continue to play substantial but comparatively smaller roles. The growth of publications from countries such as Saudi Arabia, Malaysia, Italy, and South Korea further demonstrates the broadening international engagement in this research domain.

5 Conclusion

This study examined the development and diffusion of research at the intersection of digital finance and machine learning over the past fifteen years using a bibliometric approach based on Scopus metadata. The analysis provided three main contributions. First, the temporal assessment revealed a strong increase in publication activity beginning around 2018. Second, the geospatial analysis showed clear evidence of global diffusion, with Asia—particularly India and China—emerging as dominant research hubs, while contributions from Europe, North America, and the Middle East remain substantial. Third, the keyword-based thematic analysis identified consistent growth in topics such as machine learning, deep learning, blockchain, and decentralized finance, illustrating the deepening integration of AI-driven methods and distributed technologies within financial research.

Despite these insights, several limitations must be acknowledged. The study relies on a single bibliographic source (Scopus) and employs a targeted query designed to maximize precision rather than recall. As a result, some relevant publications may not be captured, particularly those that discuss digital finance or machine learning without using explicit keywords. The use of publication metadata also limits interpretability to descriptive trends as deeper content-based insights would require full-text or abstract-level analysis.

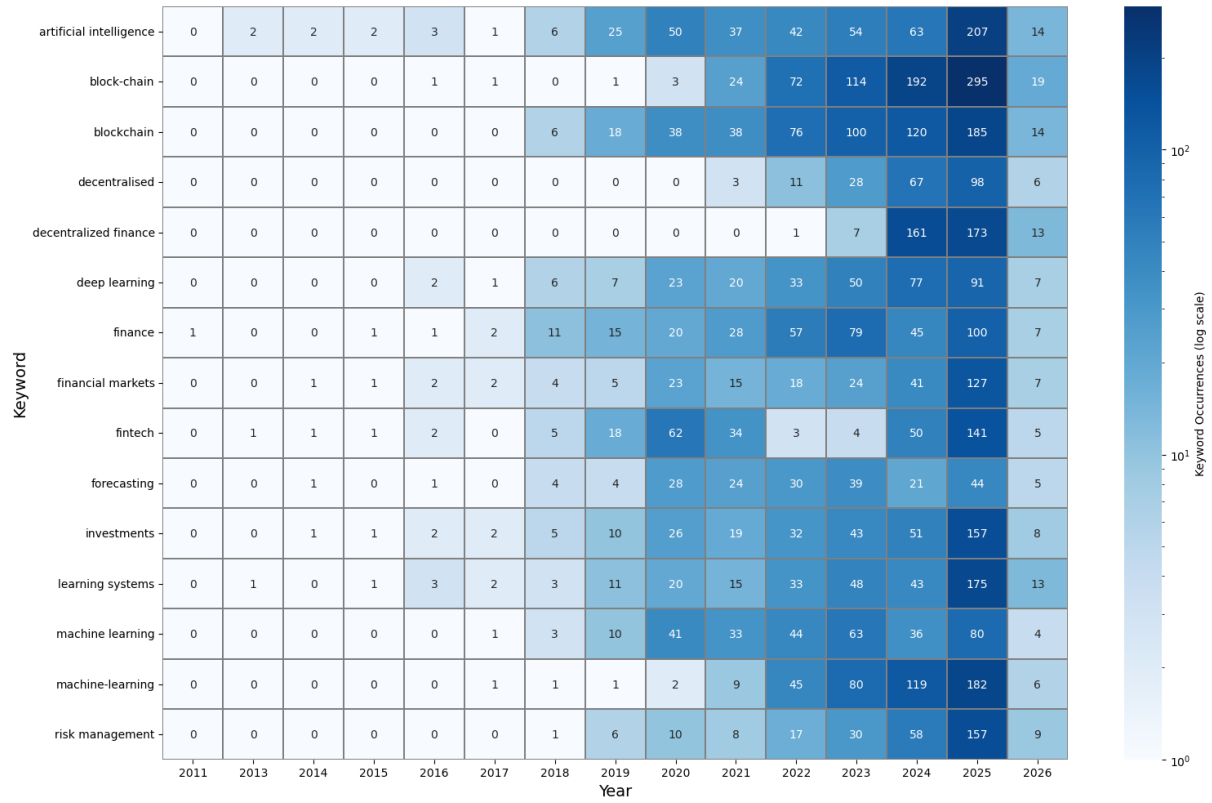


Figure 4: Heatmap of yearly occurrences for the top 15 keywords in the dataset (log scale). The figure illustrates the temporal evolution of core themes in research at the intersection of digital finance and machine learning between 2011 and 2026. Keywords associated with artificial intelligence and machine learning (such as *machine learning*, *deep learning*, and *artificial intelligence*) show a marked rise after 2018, reflecting the growing integration of advanced computational methods into financial applications. Terms related to blockchain-based technologies (including *blockchain* and *decentralized finance*) exhibit rapid growth from 2019 onward, peaking around 2024–2025 in line with global interest in distributed ledger systems. Traditional finance-related keywords (e.g., *financial markets*, *risk management*) also become more prominent over time, indicating the increasing application of machine learning techniques within established financial domains.

Future research can expand on this work in several ways. Broader data collection strategies, including merging Scopus and Web of Science queries or refining keyword taxonomies, could improve coverage and robustness. Topic modeling or embedding-based semantic analyses may uncover conceptual structures that go beyond surface-level keyword frequencies. Additionally, incorporating citation networks or collaboration networks could reveal influence dynamics and research communities within the field. Finally, extending the temporal horizon as more data becomes available would allow for monitoring whether current growth patterns stabilize, accelerate, or fragment into subfields.

Overall, this study provides an initial empirical map of how digital finance and machine learning research has grown, spread globally, and shifted in thematic focus. It offers a foundation for more detailed bibliometric or conceptual investigations and contributes to a broader understanding of how emerging technologies continue to reshape the financial research landscape.

Data Availability Statement

The bibliometric dataset and all analysis scripts used in this study are openly available on GitHub at: https://github.com/MSCA-DN-Digital-Finance/WP3_eXplainable_and_Fair_AI/tree/main/IRP1/Code/historical_finance_project.

The repository contains the Scopus extraction (in BibTeX format) and a Jupyter notebook used for data processing and the generation of all figures.

Author Contribution and AI Usage Statement

The author confirms that all analysis, coding, data processing, and interpretation of results in this report were conducted independently. The bibliometric dataset was retrieved, cleaned, and analysed by the author, and all figures were generated using Python.

Large Language Models (LLMs), specifically OpenAI's ChatGPT, were used as a writing assistant to support the formulation of text, improve clarity and structure, and provide suggestions for methodological descriptions. No AI system generated original analytical content, research results, or conclusions. All substantive analytical decisions, coding procedures, and interpretations were made by the author.

The author assumes full responsibility for the correctness, originality, and integrity of all content presented in this report.

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A Scopus Search Query

This appendix reproduces the exact Boolean search query used to retrieve the bibliometric dataset from Scopus. Line breaks were inserted manually for readability. The query was applied to the title, abstract, and author keywords fields, restricted to English-language journal articles and conference papers published between 2011 and 2026.

```
TITLE-ABS-KEY(  
  ("machine learning" OR "deep learning" OR  
   "neural network*" OR "artificial intelligence")  
  AND  
  ("fintech" OR "digital finance" OR "mobile banking" OR  
   "digital payment*" OR "blockchain" OR "cryptocurrency" OR  
   "crypto" OR "DeFi")  
  AND  
  (finance OR bank* OR "credit risk" OR "risk management" OR  
   "asset pricing" OR "portfolio")  
)
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